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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	AlgorithmX
Selected Domain	Domain 1
Selected SDG Track (1–17)	The main SDG is 4, and the most important secondary SDG is 8
Project / Solution Name	AI Skill Navigator
Name (Leader)	CHEN ZANXU
Email & Phone (Leader)	chenzanxu430@gmail.com 601135189975
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing the tasks within jobs, leaving early-career workers with a practical question: which parts of their role may change, which nearby roles fit their existing skills, and what should they learn next? Our initial focus is final-year students and workers in their first three years in Malaysia's e-commerce operations. These users may rely on routine digital tasks yet lack a clear bridge from those tasks to emerging digital roles. The World Economic Forum's Future of Jobs Report 2025 estimates that 39% of workers' core skills will change by 2030 and that 59% of workers will need training. General career advice often stops at broad job or skill lists; it rarely links a person's daily tasks to dated labour-market evidence, an explainable role match, a specific skill gap, and an affordable learning sequence. Career coaching and intensive bootcamps may also be out of reach for people with limited time or money. Without accessible, source-linked guidance, workers can struggle to choose what to learn and where to move. This creates a risk that people with less access to Career support will be less prepared for workplace change.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AI Skill Navigator is a free, no-login web prototype for final-year students and early-Skill workers (0–3 years) in Malaysia, initially focused on e-commerce operations. Users enter their role, daily tasks, experience, interests and skills, then select a human-centred or automation-first scenario. The prototype turns this profile into task-level impact estimates, adjacent-role matches with transparent skill-overlap scores, a prioritised skills gap, and a week-by-week learning plan with course links and estimated hours. Its trend layer compares dated, manually reviewed JobStreet Malaysia posting snapshots with published sources including the WEF Future of Jobs Report 2025, DOSM labour statistics, and Malaysia's Digital Economy Blueprint. Dates, source labels and small sample counts are shown so users can judge the

evidence. Scores are directional estimates, not hiring probabilities. The approach supports SDG 4.3 and 4.4 through lifelong learning and job-relevant skills, with a secondary link to SDG 8.5 through pathways toward productive work. In Phase 2, we plan to test recommendations with target users and widen role coverage.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

AI Skill Navigator is a web application built with Next.js, React and TypeScript. It is hosted and deployed on Replit. The prototype does not call an external GenAI model or third-party runtime API. Instead, it uses transparent, rule-based analysis with curated, dated trend snapshots. Based on users' roles, tasks, experience, interests and skills, it produces task-level impact estimates, adjacent-role matches, skill gaps and a prioritised learning plan.

The system flow is: user profile → curated trend data → in-app rule-based analysis and role matching → skill gaps and learning plan → results.

Official SDC sponsor-provided resource used: Yes — Replit was used as the development and cloud deployment platform to host the prototype. No sponsor-provided GenAI API or external AI model was used. RunPod was not used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We used a focused design-thinking approach. We narrowed the initial user group to students and workers in their first three years in e-commerce operations, then mapped their needs into a three-step flow: profile current work, choose an AI-adoption scenario, and review a `career` roadmap. To keep the prototype feasible, we started with a limited set of adjacent roles and dated, manually reviewed evidence rather than claiming full labour-market coverage. Visible source dates, sample counts, illustrative labels and skill-match factors help users understand how recommendations were formed and where evidence is limited. The app

is deployed on Replit and can be tried without login. Before Phase 2, we plan task-based sessions with target users, track completion and comprehension, assess whether role and course suggestions are useful, and revise the flow and data.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG Target 4.4 — Increase the number of youth and adults with relevant skills for employment, decent jobs and entrepreneurship. The project helps users identify relevant skill gaps and plan learning actions. Secondary: SDG Target 8.5 — Support young people and other workers in navigating pathways toward productive employment and decent work. The application provides Skill-transition guidance; it does not guarantee employment.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The World Economic Forum's Future of Jobs Report 2025 estimates that 39% of workers' core skills will change by 2030, and that 59 out of every 100 workers will need training by 2030. These are global workforce estimates, not measurements of our users or of Malaysia alone. Our own user-level baseline has not yet been measured and will be collected before the Phase 2 pilot.
Core Measurable KPI *The specific unit you will track*	Percentage-point change from pre-use to post-use in the share of pilot participants who can identify at least three relevant skill gaps and one concrete next learning action. Measure using the same short questionnaire before and after using the application.

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed pilot target: recruit at least 20 intended users and achieve a minimum 20-percentage-point increase in participants who can identify at least three relevant skill gaps and one next learning action after using the application. This is a target for evaluation, not a result already achieved.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

[Date]10/24/2026